

Esp32Tap Breadboard — Point-to-Point Wiring List

Each row is one physical connection. Map the RJ45 breakout terminals and SOT-23 adapters with a meter before using their labels.

Start with all power disconnected. Leave STANDALONE POWER and COIL POWER removed. The relay datasheet pin drawing is a bottom view—meter-verify K1 before wiring.

A. RJ45 pass-through and relay contacts

#	From	To	Part	Note
1	CONSOLE RJ45 pin 1	GND	Jumper wire	First ground
2	MOTOR RJ45 pin 1	GND	Jumper wire	First ground
3	CONSOLE RJ45 pin 7	GND	Jumper wire	Second ground
4	MOTOR RJ45 pin 7	GND	Jumper wire	Second ground
5	CONSOLE RJ45 pin 2	+8V_RAW	Jumper wire	First +8 V contact
6	MOTOR RJ45 pin 2	+8V_RAW	Jumper wire	First +8 V contact
7	CONSOLE RJ45 pin 8	+8V_RAW	Jumper wire	Second +8 V contact
8	MOTOR RJ45 pin 8	+8V_RAW	Jumper wire	Second +8 V contact
9	CONSOLE RJ45 pin 3	MOTOR RJ45 pin 3	Jumper wire	Direct pass-through
10	CONSOLE RJ45 pin 4	MOTOR RJ45 pin 4	Jumper wire	Direct pass-through
11	CONSOLE RJ45 pin 5	MOTOR RJ45 pin 5	Jumper wire	Direct safety pass-through
12	CONSOLE RJ45 pin 6	K1 G5V-2 pin 4	Jumper wire	Relay NC contact
13	MOTOR RJ45 pin 6	K1 G5V-2 pin 6	Jumper wire	Relay COM contact
14	K1 G5V-2 pin 8	R_TX 100 Ω , end 2	Jumper wire	Relay NO / ESP transmit

B. Local power

#	From	To	Part	Note
15	+8V_RAW	RXEF075 lead 1	Wire	Local electronics tap only
16	RXEF075 lead 2	1N5822 anode	Wire	Non-striped diode end
17	1N5822 cathode	VIN	Wire	Striped diode end
18	VIN	P6KE12A cathode	Wire	Striped TVS end
19	P6KE12A anode	GND	Wire	TVS return
20	VIN	TSR1-2433E pin 1	Wire	Converter input
21	TSR1-2433E pin 2	GND	Wire	Converter ground
22	TSR1-2433E pin 3	TSR_3V3	Wire	Converter output
23	VIN	22 μ F positive	Wire	Bulk capacitor
24	22 μ F negative	GND	Wire	Observe polarity
25	VIN	10 μ F input-cap positive	Wire	Near TSR input
26	10 μ F input-cap negative	GND	Wire	Observe polarity
27	TSR_3V3	10 μ F output-cap positive	Wire	Near TSR output
28	10 μ F output-cap negative	GND	Wire	Observe polarity
29	DevKit 3V3	LOGIC_3V3	Jumper wire	Logic rail
30	DevKit GND	GND	Jumper wire	Common ground
31	TSR_3V3	LOGIC_3V3	STANDALONE POWER jumper	Leave removed when USB is attached

C. TPS3700 voltage monitor

#	From	To	Part	Note
32	VIN	TPS3700 pin 5 VDD	Wire	Supply
33	TPS3700 pin 2	GND	Wire	Ground
34	VIN	R_UV_TOP 150 kΩ, end 1	Wire	UV divider
35	R_UV_TOP end 2	UV_SENSE	Wire	Divider midpoint
36	UV_SENSE	TPS3700 pin 3 INA+	Wire	UV input
37	UV_SENSE	R_UV_BOTTOM 10 kΩ, end 1	Wire	Divider bottom
38	R_UV_BOTTOM end 2	GND	Wire	Divider return
39	UV_SENSE	C_UV 1 nF, end 1	Wire	Filter
40	C_UV end 2	GND	Wire	Filter return
41	VIN	R_OV_TOP 255 kΩ, end 1	Wire	OV divider
42	R_OV_TOP end 2	OV_SENSE	Wire	Divider midpoint
43	OV_SENSE	TPS3700 pin 4 INB-	Wire	OV input
44	OV_SENSE	R_OV_BOTTOM 10 kΩ, end 1	Wire	Divider bottom
45	R_OV_BOTTOM end 2	GND	Wire	Divider return
46	OV_SENSE	C_OV 1 nF, end 1	Wire	Filter
47	C_OV end 2	GND	Wire	Filter return
48	TPS3700 pin 1 OUTA	TREAD_OK	Wire	Open-drain output
49	TPS3700 pin 6 OUTB	TREAD_OK	Wire	Tie outputs together
50	LOGIC_3V3	R_TREAD_PULLUP 10 kΩ, end 1	Wire	Pull-up
51	R_TREAD_PULLUP end 2	TREAD_OK	Wire	Pull-up node
52	TREAD_OK	R_TREAD_PULLDOWN 100 kΩ, end 1	Wire	Power-off pull-down
53	R_TREAD_PULLDOWN end 2	GND	Wire	Pull-down return
54	TREAD_OK	R_TREAD_MCU 4.7 kΩ, end 1	Wire	GPIO isolation
55	R_TREAD_MCU end 2	DevKit GPIO6	Wire	TREAD_OK_MCU
56	VIN	C_TPS3700 100 nF, end 1	Wire	Decoupling
57	C_TPS3700 end 2	GND	Wire	Decoupling return

D. SN74AHC08N gates

#	From	To	Part	Note
58	LOGIC_3V3	AHC08 pin 14	Wire	VCC
59	AHC08 pin 7	GND	Wire	Ground
60	DevKit GPIO21	AHC08 pin 1	Wire	RELAY_CMD
61	DevKit GPIO21	R_RELAY_CMD_PD 10 kΩ, end 1	Wire	Command pull-down
62	R_RELAY_CMD_PD end 2	GND	Wire	Pull-down return
63	TREAD_OK	AHC08 pin 2	Wire	Relay permission
64	AHC08 pin 3	RELAY_GATE	Wire	Relay gate output
65	DevKit GPIO15	AHC08 pin 4	Wire	TX_ENABLE
66	DevKit GPIO15	R_TX_ENABLE_PD 10 kΩ, end 1	Wire	Enable pull-down
67	R_TX_ENABLE_PD end 2	GND	Wire	Pull-down return
68	TREAD_OK	AHC08 pin 5	Wire	TX permission

#	From	To	Part	Note
69	AHC08 pin 6	TX_GATE	Wire	TX gate output
70	AHC08 pins 9,10,12,13	GND	Four wires	Tie unused inputs low
71	AHC08 pins 8,11	NO CONNECTION	—	Unused outputs
72	AHC08 pin 14	C_AHC08 100 nF, end 1	Wire	Place close to IC
73	C_AHC08 end 2	AHC08 pin 7	Wire	Decoupling return

E. TPS709, BC337, and relay coil

#	From	To	Part	Note
74	VIN	TPS709 pin 1 IN	Wire	Regulator input
75	TPS709 pin 2	GND	Wire	Ground
76	RELAY_GATE	TPS709 pin 3 EN	Wire	Hardware-gated enable
77	TPS709 pin 4	NO CONNECTION	—	NC pin
78	TPS709 pin 5 OUT	+5V_RLY	Wire	Relay supply
79	VIN	C_TPS709_IN 1 μ F, end 1	Wire	Input bypass
80	C_TPS709_IN end 2	GND	Wire	Bypass return
81	+5V_RLY	C_TPS709_OUT 4.7 μ F positive	Wire	Output capacitor
82	C_TPS709_OUT negative	GND	Wire	Output return
83	RELAY_GATE	R_BASE 560 Ω , end 1	Wire	Base resistor
84	R_BASE end 2	BC337 base	Wire	Verify physical lead
85	BC337 base	R_BASE_PD 10 k Ω , end 1	Wire	Base pull-down
86	R_BASE_PD end 2	GND	Wire	Pull-down return
87	BC337 emitter	GND	Wire	Verify physical lead
88	BC337 collector	K1 pin 16	Wire	RELAY_COIL–
89	+5V_RLY	K1 pin 1	COIL POWER jumper	Leave removed initially
90	K1 pin 1	P6KE6.8CA lead 1	Wire	Bidirectional coil TVS
91	P6KE6.8CA lead 2	K1 pin 16	Wire	Bidirectional coil TVS

F. SN74AHC126N and UART

#	From	To	Part	Note
92	LOGIC_3V3	AHC126 pin 14	Wire	VCC
93	AHC126 pin 7	GND	Wire	Ground
94	TX_GATE	AHC126 pin 1	Wire	Gate 1 OE
95	DevKit GPIO17	AHC126 pin 2	Wire	Gate 1 input
96	AHC126 pin 3	R_TX 100 Ω , end 1	Wire	Gate 1 output
97	AHC126 pins 4,5,9,10,12,13	GND	Six wires	Unused enables/inputs low
98	AHC126 pins 6,8,11	NO CONNECTION	—	Unused outputs
99	AHC126 pin 14	C_AHC126 100 nF, end 1	Wire	Place close to IC
100	C_AHC126 end 2	AHC126 pin 7	Wire	Decoupling return
101	CONSOLE RJ45 pin 6	R_CONS_RX 10 k Ω , end 1	Wire	Console RX tap
102	R_CONS_RX end 2	DevKit GPIO18	Wire	Console RX
103	RJ45 pin-3 net	R_PIN3_RX 10 k Ω , end 1	Wire	Pin-3 tap

#	From	To	Part	Note
104	R_PIN3_RX end 2	DevKit GPIO16	Wire	Pin-3 RX

G. Relay feedback, USB presence, and LEDs

#	From	To	Part	Note
105	K1 pin 11	GND	Wire	Feedback COM
106	K1 pin 13	DevKit GPIO4	Wire	NC feedback
107	LOGIC_3V3	R_FB_NC 10 kΩ, end 1	Wire	NC pull-up
108	R_FB_NC end 2	DevKit GPIO4	Wire	NC pull-up node
109	K1 pin 9	DevKit GPIO5	Wire	NO feedback
110	LOGIC_3V3	R_FB_NO 10 kΩ, end 1	Wire	NO pull-up
111	R_FB_NO end 2	DevKit GPIO5	Wire	NO pull-up node
112	DevKit 5V/VBUS	2N7000 gate	Wire	Verify physical lead
113	DevKit 5V/VBUS	R_VBUS_DISCHARGE 10 kΩ, end 1	Wire	Gate discharge
114	R_VBUS_DISCHARGE end 2	GND	Wire	Discharge return
115	2N7000 source	GND	Wire	Verify physical lead
116	2N7000 drain	DevKit GPIO7	Wire	VBUS_PRESENT_N
117	LOGIC_3V3	R_VBUS_PULLUP 10 kΩ, end 1	Wire	Drain pull-up
118	R_VBUS_PULLUP end 2	DevKit GPIO7	Wire	Pull-up node
119	DevKit GPIO38	R_STATUS 330 Ω, end 1	Wire	Status LED
120	R_STATUS end 2	Green LED anode	Wire	Long LED lead
121	Green LED cathode	GND	Wire	Short/flat-side lead
122	LOGIC_3V3	R_POWER_LED 1 kΩ, end 1	Wire	Power LED
123	R_POWER_LED end 2	Red LED anode	Wire	Long LED lead
124	Red LED cathode	GND	Wire	Short/flat-side lead
125	LOGIC_3V3	C_LOGIC 100 nF, end 1	Wire	Logic rail bypass
126	C_LOGIC end 2	GND	Wire	Bypass return

H. Before power

Check	Expected result
K1 pins 4 to 6	Continuity with COIL POWER removed
K1 pins 8 to 6	Open with COIL POWER removed
+8V_RAW to GND	No short
VIN to GND	No short; capacitor charging may briefly affect reading
LOGIC_3V3 to GND	No short
STANDALONE POWER	Removed while UART USB is connected
COIL POWER	Removed until the relay supply and driver are bench-tested